Round 7 | Three moment corrections

Fixed-weight ReLU means: variance, fourth cumulants and sampling | 23 September 2026

My first experiment is a noise-aware audit of the covariance closure, followed by a specific, almost-free term patch. My first new κ_4 representation is a quartic response polynomial, evaluated afresh at every ReLU and matched to the reliable diagonal/(2,2) slices. Neither proposal removes the old K3 tier. Neither has demonstrated your target accuracy yet.

Derived distinction. Three marginal numbers determine the local mean correction; they do not form a closed propagation state. Their future values still depend on joint cumulants. Your oracles establish where to intervene, not that off-diagonal information can be discarded.

Candidate	Additional network cost	Decision gate
Covariance terms absent from inspected V29	<0.2 u; existing covariance transport reused	Test at amplitude one; reject if projected variance error grows.
Fresh κ_4 kernel, diagonal and (2,2) matched	About 26 u for all outputs with existing Strassen	Needs >9.1% raw-MSE reduction at that cost.
Same kernel also supplies full next (3,1) slice	About 42 u total, not 26+42	Needs >13.9% raw-MSE reduction; can affect both channels.
Add the full K3×C directional response	About 54 u extra for 350 queries/layer	Only after the cheaper kernel passes; count actual ranks.
10,000 ordinary forward samples	156.25 u + moment statistics; CV costs extra	Needs >37.5% raw-MSE reduction before CV overhead.

Evidence labels. DERIVED means algebra or an explicit cost calculation. CODE-AUDITED refers to the inspected public V29. CONJECTURE means accuracy is unknown. MEASURE-THIS specifies a falsifier. I did not receive your teacher arrays; the tests run here check algebra, not leaderboard accuracy.

Conventions and scope

Let W have next-layer weight vectors as columns, so $z' = W^T h$. C is the current pre-ReLU covariance; $X = z - \mu$; $\sigma^2 = \mathrm{diag}(C)$; $\alpha = \mu/\sigma$. Write K3 and K4 for ordinary symmetric cumulant tensors, without source-format normalization factors. Use the actual complete K3 implementation, including AAP, MPP and thin terms, when evaluating it.

Accounting: 1 u = 2^{31} FLOPs. A float32 $(n \times n)(n \times m)$ product costs $nm(2n-1)/2^{31} \approx m/n$ u. Dense products use 0.51 u only where the existing Strassen routine applies; rectangular estimates below are classical. Float64 doubles costs. Exactly 260/1024 = 0.2539B, not 0.244B; substitute 250 if that is the actual measured baseline.

1 / Separate teacher noise from closure error

Derived. With N independent samples, the leading sampling variance of a centered covariance estimate is

$$\mathrm{Var}(\widehat{C}_{ij}) = \frac{K_{iijj} + C_{ii}C_{jj} + C_{ij}^2}{N} + O(N^{-2}).$$

At N=6×10⁷ and typical post-ReLU variance 0.30, weakly correlated Gaussian-like entries have SE≈3.87×10⁻⁵. Thus 4.2×10⁻⁵ per entry could contain substantial teacher noise. Insert your actual C and K22 rather than assuming 0.30. This observation does *not* dismiss the variance oracle.

$$\frac{\mathrm{SE}(\hat{v})}{v} = \sqrt{\frac{2+\gamma_2}{N}}, \qquad \mathrm{SE}(\hat{\kappa}_4/v^2) \simeq \sqrt{24/N}.$$

For nearly Gaussian preactivations these numbers are **0.0183%** for variance and **6.32×10⁻⁴** for standardized κ₄. The observed 0.15% variance error is much larger than this variance noise. But if excess kurtosis is 0.003, the κ₄ teacher itself has roughly 21% relative sampling error. Report noise-corrected pattern error before interpreting “20% κ₄ error.”

State/cost/falsifier. Use the stored post C and K22: O(n²) state, <0.02 u/layer for an entrywise noise map. Compute the projected variance noise from the stored next-preactivation v and κ₄. If independent teacher blocks exist, with residuals R_A and R_B against the same deterministic chain, use ⟨R_A,R_B⟩ to estimate squared systematic error. For a fresh residual that subtracts inherited teacher error, pre/post noise is correlated: subtracting two independent noise variances is wrong.

An exact fresh-residual noise audit requires paired block estimates or cross pre/post fourth moments, which your listed aggregate arrays do not include. Without them, report the entrywise noise scale and leave the fresh-error decomposition unresolved. No new 60-million-sample run is proposed.

Why querying 350 variances does not close the state

Once C_post is known, m quadratic forms cost one C_postW_m product plus column dot products: m/n≈0.342 u at m=350. V29 still needs C for births and other closures, so this is not automatically a saving. A correction matrix can instead be added before the existing transport, with no new dense product.

$$\mathbb{E}_w[(w^T R w)^2] = \frac{4}{n^2}\left[(\mathrm{tr}\,R)^2 + 2\|R\|_F^2\right].$$

For omitted off-diagonal covariance R, the trace term vanishes. With flat error energy, retaining a fraction p of pairs leaves relative projection RMS √(1−p). Even reducing 0.15% to 0.10% needs p≥0.556 in this model. “A few dominant pairs” needs a measured energy concentration; an adjoint does not create it.

2 / A concrete covariance patch before a new state

Code-audited [1]. The inspected V29 (1,1) table contains the Gaussian C and C² terms, and a standalone wκ431 term. It omits the pure Gaussian C³/C⁴ terms and the C×κ31 term below. If your current branch already contains them, skip this proposal.

$$d_1 = \Phi(\alpha), \quad d_2 = \phi(\alpha)/\sigma, \quad d_3 = -\alpha\phi(\alpha)/\sigma^2, \quad d_4 = (\alpha^2 - 1)\phi(\alpha)/\sigma^3.$$

Derived [2]. Gaussian covariance is $\sum_{r\geq 1} C_{ij}^{\,r} d_{r,i} d_{r,j}/r!$. First-order bivariate Edgeworth dressing of κ31 is $(\kappa_{iiij}/6)\sum_{r\geq 0} C_{ij}^{\,r} d_{r+3,i} d_{r+1,j}/r!$ plus transpose. Therefore a specific missing correction is

$$\Delta C_{ij} = \frac{C_{ij}^3}{6} d_{3,i} d_{3,j} + \frac{C_{ij}^4}{24} d_{4,i} d_{4,j} + \frac{C_{ij}}{6}\left(d_{4,i}\kappa_{iiij} d_{2,j} + d_{2,i}\kappa_{ijjj} d_{4,j}\right), \qquad i \neq j.$$

Set its diagonal to zero. This extends the existing K3 closure; it is not a replacement by marginal Gaussian propagation. The earlier exact-Mehler-on-K2 result did not test this incremental correction after the cross-neuron K3 channel was retained. It could nevertheless be too small, already compensated elsewhere, or detrimental.

State/update/cost. One n×n correction buffer (4 MiB), existing C, κ31 and Wick coefficients. Compute the displayed Hadamard terms, then C_post←C_post+ΔC before the existing transport. A conservative $24n^2+50n$ arithmetic count is <0.012 u/layer, <0.2 u/network. No source transport or new matrix product is needed in deployment.

First falsifier on the stored arrays

At layers 8, 12 and 15: (i) teacher-force the bivariate closure with all available true preactivation cumulants; (ii) separately replace only κ31, only K22 and only D21; (iii) test the three displayed terms at their derived coefficients, without fitting a multiplier. Project each covariance residual through the actual next W onto the selected neurons. A projection costs 0.342 u for 350 queries; closure work is $O(n^2)$. Cap this diagnostic at 10 u total.

Reject the patch if it fails to reduce the noise-adjusted, sensitivity-weighted projected error on held-out layers. If teacher-forcing κ31 removes the systematic variance error, the priority is better κ31, not a new covariance representation. If even fully teacher-forced closure fails, fourth-order bivariate data may be insufficient: fifth/sixth mixed cumulants and higher Edgeworth products remain possibilities, not identified causes.

End-to-end. First inject only the resulting variance correction into the mean call, matching your oracle experiment. Then test the actual covariance-state patch with all downstream feedback enabled and bill it. Keep it only if final adjusted MSE improves on held-out networks. The single-network oracle percentages are ceilings for these interventions, not promised gains.

3 / A quartic response regenerated at each ReLU

Derived connected-diagram truncation [3]. Expand the centered activation in marginal Gaussian Wick polynomials: h_j−E_Gh_j≈d₁,jX_j+(d₂,j/2):X_j²:+(d₃,j/6):X_j³:. For a query w define

$$u = d_1 \circ w, \quad b = d_2 \circ w, \quad c = d_3 \circ w, \quad q = Cu, \quad t = b \circ q.$$

The connected four-output trees contract to the following homogeneous quartic polynomial:

$$F(w) = K_4[u,u,u,u] + 12K_3[t,u,u] + 4\sum_j c_j q_j^3 + 12t^T Ct.$$

The four terms are: one input fourth-cumulant edge; a third-cumulant edge joined to a covariance edge; a Gaussian three-edge star; and a Gaussian three-edge path. The coefficients follow from cumulant multilinearity and connected partitions. For example, 4 cum(L,L,L,Q)=12∑b_jq_jK3(j,u,u)+2∑b_jK5(j,j,u,u,u), with the latter omitted here.

This is a fresh nonlinear response at every layer, not a star born once and transported linearly. It retains the signed weights and actual realized K3 queries. No averaging over later W is performed. Its all-distinct entries include the leading κ₃×C and κ₄ structures requested.

A first version with no new source contractions

Temporarily omit the K3 term and use the current harmonic fourth-cumulant approximation K4=Sym(G⊗ηI), G=λC+diag(g−λ diag C). Then

$$F_0(w) = \eta\|u\|^2 u^l Gu + 4c^l q^{\circ 3} + 12t^l Ct.$$

Gu=λq+(g−λ diag C)∘u is elementwise after q is available. Thus all m raw diagonal queries need only CU and CT: **2m/n u/layer**, plus O(nm). This cheap version tests the missing path and local regeneration; it does not claim to recover the full K3-driven pattern.

What this approximation loses

F is not the full fourth cumulant of even the truncated polynomial. It omits cycles, higher marginal Hermite orders, input K5/K6, and additional K4×C and K3² terms. For Gaussian L+Q alone, the exact additional ring is 3 tr[(C diag b)⁴]. A generic evaluation costs O(n³) per query, so it cannot be called free. Weak off-diagonal correlations justify a diagram ordering for distinct indices, not a uniform 2% guarantee, especially for repeated indices.

The next page matches reliable repeated-index slices exactly within the surrogate. The appendix gives the leading repeated-index law and inexpensive harmonic K4×C contractions. All accuracy claims beyond the derived terms are **conjectures**.

4 / Preserve the accurate pieces; test the missing pattern

Derived. Let F_0 define a symmetric tensor H by $F_0(w)=H[w,w,w,w]$. Its diagonal and (2,2)/(3,1) slices are available using only C_ij, C_ii, G_ij and the local coefficients: $O(n^2)$, without forming H. Match the reliable post-ReLU diagonal d and off-diagonal K22 by

$$\widetilde{F}(w) = F_0(w) + \sum_j \delta d_j w_j^4 + 3(w^{\circ 2})^T \Delta_{22} w^{\circ 2},$$

where $\delta d = d - \mathrm{diag}(H)$, $\Delta 22 = K22 - H22$ and $\mathrm{diag}(\Delta 22)=0$. This avoids adding a new estimate of the already-correct level on top of the old estimate. It replaces the next κ_4 diagonal prediction, rather than adding a full cumulant to λC regeneration.

State/update/cost. Reuse C and the current harmonic G parameters; keep Δ22 (4 MiB) and O(nm) query buffers. Each layer recomputes this polynomial and reads out the actual next-layer queries. Three rectangular products CU, CT and $\Delta 22(W \circ W)$ cost **3m/n u/layer**: 1.025 u for m=350. For all n outputs, 3×0.51×15=22.95 u in products; allow about **26 u/network** including the explicit $O(n^2)$ algebra. Count the realized mask size; the κ_4-relevant mask may be larger than 350.

The same polynomial can supply the (3,1) slice

$$H[w_i, w_i, w_i, w_k] = \tfrac{1}{4} \nabla \widetilde{F}(w_i)^T w_k.$$

For the Gaussian part, r=Ct and

$$\nabla F_G = 4d_3 \circ q^3 + 24 d_2 \circ q \circ r + d_1 \circ C(12c \circ q^2 + 24 b \circ r).$$

The harmonic gradient uses the existing q; matching adds $4\delta d \circ w^3 + 12 w \circ (\Delta 22 w^2)$. At all n queries, CU, CT, the displayed reverse C product, $\Delta 22 W^2$ and the final gradient^{T}W are **five** dense products/layer: 38.25 u over 15 layers with Strassen; allow about **42 u total**. This is the total for diagonal plus κ31. It is not the previously dead adjoint scheme for saving old-source contractions.

Falsifier and deployment test

At one late layer, form the true pair-only quartic $P_{true}(w) = \sum d_j w_j^4 + 4 \sum_{j \neq k} K31_{jk} w_j^3 w_k + 3 \sum_{j \neq k} K22_{jk} w_j^2 w_k^2$. Compare $F_0 - P_H$ with $\kappa 4_{true,next} - P_{true}$; this isolates the (2,1,1)/(1,1,1,1) prediction. Full pair matching here is an **oracle diagnostic**; deployment above only uses the existing predicted d and K22. Also test its uncorrected κ31 prediction directly. One layer with 350 queries costs <3 u in principal products plus $O(n^2)$ terms; use a 5-u cap.

Reject if the sensitivity-weighted residual MSE does not improve, or if the end-to-end diagonal-only replacement fails the 9.1% raw-MSE gate for a 26-u bill. Enable κ31 feedback into covariance only after its own held-out slice test; the 42-u version needs >13.9% raw improvement. Do not inject teacher K31 into a submitted estimator.

5 / Price the genuinely new K3×C response

Derived. This uses existing K3, without changing its transport law. For the illustrative source K_s=Sym∑p_r⊗p_r⊗a_r, put z=P^Tu and y=A^Tu. Then

$$K_s[t,u,u] = \tfrac{1}{3}\left[(z\circ z)^T(A^Tt) + 2(z\circ y)^T(P^Tt)\right].$$

The next-layer leg transport already computes P^Tu and A^Tu because u=diag(Φ)w. Reuse these arrays. The extra work is A^TT and P^TT for T=[t_1,...,t_m]. Apply the corresponding derivative of the complete source cubic for AAP and thin terms; do not silently replace V29 by an MPP-only model.

$$\Delta c_\ell \simeq 2\tfrac{m}{n}\sum_{s\ \text{live}} \rho_s + \tfrac{m}{n}\tfrac{r_c}{n}\quad \text{u}, \qquad \rho_s = 1\ \text{(dense)},\quad r_s/n\ \text{(factored)}.$$

For old sources, form Q^TT once and multiply by the stored source factors; no Hadamard product of reconstructed legs is needed. The extra query buffers are O(nm+r_cm), only a few MiB; preserve the current-layer factors until the query is finished. Thin-leg products and coefficients must also be metered.

Using the supplied 120 source-layer schedule as a conservative ledger, ranks n through age 4, 384 through age 7, and 224 thereafter give ∑ρ=73.125. At m=350 the two projected legs cost **49.99 u/network**, plus about 2 u shared-basis projections and thin work: plan roughly **54 u extra**. If reuse of the u projections is unavailable, add their cost; it is not a 54-u implementation. At m=n, the two new projections alone cost 146.25 u, beyond the κ_4-only oracle headroom.

The query is new information: it depends on off-slice K3 and on the curvature direction t=(d_2∘w)∘C(d_1∘w), rather than on the 13 already-tested local features. It also differs from transporting a fourth-cumulant star across depth.

First falsifier before paying the full bill

Capture the pre-ReLU source factors and the reused next-layer legs at layer 12. Evaluate the added term for 32 sensitive output rows, then 350 only if it helps. Recompute the diagonal/(2,2) matching for the complete F. This costs only O(n^2): restricting w to one or two coordinates restricts every K3 index in K3[t,u,u] to those coordinates, so D3 and D21 suffice for the matching terms. A blind additive update would double-count known pair content. Cap the initial query test at 2 u and the expanded layer at 10 u.

End-to-end gate. A selected-query kernel with this extension will be about 70-80 u above the baseline, depending on thin legs and matching. A 75-u bill requires >22.4% raw-MSE reduction. That is roughly three quarters of the measured κ_4-only oracle benefit. I would not build it before the cheap kernel and the one-layer query have shown a substantial pattern gain.

The 54-u figure prices the new directional queries; adding the selected-query, pair-matched base costs about another 18 u. This is a complete specified polynomial truncation, not a complete fourth-cumulant closure. Reject the implementation if thin-leg work or lost reuse pushes it past its declared bill.

6 / Sampling corrections: subtracting a constant is not a CV

Derived. For y=z−μ, the centered influence functions for the three cumulants are

$$\psi_2 = y^2 - v, \qquad \psi_3 = y^3 - 3vy - \kappa_3,$$

$$\psi_4 = y^4 - 6vy^2 - 4\kappa_3 y + 3v^2 - \kappa_4.$$

Estimating δv as a sample variance minus the chain value leaves the sampling variance unchanged. A usable estimator is an unbiased sample cumulant minus the chain estimate minus β times the sample mean of a zero-mean Gaussian Hermite control; fit β on an independent split. Its leading noise is governed by these influence functions. Near Gaussianity their variances are (2v², 6v³, 24v⁴). Raw moments up to eight, or paired samples, are needed to assess non-Gaussian fourth-cumulant estimator noise accurately.

For a jointly Gaussian standardized target and surrogate with correlation ρ, Hermite orthogonality gives Corr(H_q(Z),H_q(G))=ρ^q. Optimal control reduces variance by $1-\rho^{2q}$. A **100× reduction** needs |ρ|≥0.99749, 0.99833 and 0.99874 for q=2,3,4 respectively. Similar marginal distributions or accurate analytic moments do not establish such a samplewise coupling.

Ordinary samples	Full-network forward cost	Near-Gaussian relative SE of variance
10,000	156.25 u	1.414%
100,000	1,562.5 u	0.447%
2,000,000	31,250 u	0.100%

Even a 100× reduction at 10,000 samples gives variance relative SE≈0.141%, slightly above 0.1%. A perfectly batched Strassen implementation can reduce the dense-forward bill toward 0.51 of these costs, with activation/statistics work additional; it does not change the statistical sample requirement. The chain plus 100,000 classical forwards exceeds B.

One inexpensive falsifier, if a new coupling is claimed

Use 2,048 paired inputs and a fixed Gaussian tangent surrogate G_l(x)=m_l+J_lx, $J_l=W_l^{T}$ diag(Φ_{l−1})J_{l−1}. State: the running J and selected rows, O(n²), plus streaming moments. Building the 16 maps costs ≤16 u classically, actual MLP forwards 32 u, and evaluating 350 surrogate rows at each of two selected layers costs about 1.37 u. Allow <51 u including elementwise statistics. A Gaussian surrogate’s cumulants and Hermite means are analytic; fit CV coefficients on one split and test residual variance on another.

Compute the held-out residual variance of the actual ψ₂/ψ₃/ψ₄ and their mean-sensitivity-weighted combination. Reject unless the observed gain, including surrogate evaluation cost, crosses the sample-count gate. Aggregate teacher moments cannot provide this coupling test. The old 1.5× mean-CV measurement did not directly measure these influence functions; the Gaussian benchmark makes a 100× surprise unlikely.

Sampling variants: what the new diagnosis does not rescue

Derived. In the Gaussian benchmark, H_2 and H_4 are even, so antithetic sign pairs duplicate their noise; a pair does not reduce their variance at fixed evaluation cost. Averaging over a resampled last W changes the fixed-network problem. Radial Rao-Blackwellization is exact for raw moments: $E[f(X)^k]=E[R^k]E[f(\sqrt{n}U)^k]/n^{k/2}$. But $CV^2(R^k)\approx k^2/(2n)$, only 0.20% for k=2 and 0.78% for k=4. It helps dramatically only if angular variation is already tiny. Your sphere/QMC negatives stand; no new 100× mechanism follows from homogeneity.

7 / Which experiment first?

Order	Experiment and stop rule	Diagnostic cost
1. Variance	Noise map, teacher-forced bivariate closure, then the fixed-coefficient covariance patch.	<10 u for three selected layers; no new MC.
2. κ_4 diagonal	Compare the pair-subtracted quartic kernel with the true omitted (2,1,1)/(1,1,1,1) contribution.	<5 u for one late layer.
3. Coupled κ31 / variance	Only if step 2 or κ31 teacher forcing is positive: test the quartic-gradient κ31 slice.	<6 u for one full layer with existing Strassen.
4. K3×C contribution	Capture factors; test 32 curvature-direction queries, then 350.	2-u then 10-u caps for one layer.
5. Sampling	Only for a genuinely new coupling: test influence-function residuals on paired inputs.	<51 u; no promise of a viable estimator.

Planning rank: variance first, κ_4 second, κ_3 third pending the other model's transport work. The first two can interact through κ31; their oracle gains cannot simply be added. There is no defensible numerical "expected gain" before these local tests, so use the measured oracle headroom and break-even cost gates instead.

$$\frac{\text{score}_{new}}{\text{score}_{base}} = \frac{\text{MSE}_{new}}{\text{MSE}_{base}}\left(1+\frac{c}{260}\right).$$

For an intervention achieving exactly the supplied single-channel oracle error, the maximum break-even extra cost is about **228 u for variance, 112 u for κ_4, and 43 u for κ_3**. These are optimistic ceilings, not recommended budgets. A 26-u kernel needs to recover about 30% of the κ_4 oracle's MSE reduction; a 75-u extension needs about 74%.

End-to-end protocol. Fix coefficients and masks before held-out evaluation. Start with the same isolated mean-input intervention as the oracles; then enable the intended state feedback. Compare paired final errors against the existing 10^9-sample means over the available networks, report raw and adjusted MSE, and bill all extra preparation, contractions and statistics. Reject a one-network improvement that does not survive this test.

Conclusion. The strongest new cheap test is the audited covariance term patch; the strongest new state candidate is the pair-matched quartic response. Its limitation is explicit omitted diagrams, not an assumed cancellation or a new rank claim. No result here establishes the leaders' method or halves V29's cost.

Appendix A / Repeated indices and omitted κ₄ response

Derived repeated-index law. Let m_r=E_G[h_r], η_r=2m_r(1−d₁,r), and β_r=2d₁,r−2m_rd₂,r−2d₁,r². For three distinct neurons r,k,m, the leading Gaussian tree and single cross-K3 terms are

$$K^{post}_{rrkm} \approx \beta_r d_{1,k} d_{1,m} C_{rk} C_{rm} + \eta_r d_{1,k} d_{1,m} K_{rkm} + \eta_r (d_{2,k} d_{1,m} C_{rk} C_{km} + d_{1,k} d_{2,m} C_{rm} C_{km}).$$

Proof: set B_r=(h_r−m_r)²−Var_G(h_r). Then κ(h_r,h_r,h_k,h_m)=κ(B_r,h_k,h_m)−2Cov(h_r,h_k)Cov(h_r,h_m). Its local derivative averages are E[B′_r]=η_r and E[B″_r]=2d₁,r−2m_rd₂,r. This explains the star coefficient and the additional chain/K3 terms. Higher mixed cumulants and cross-products are omitted. Thus matching only diagonal/(2,2) does not fully resum repeated-index (2,1,1) terms.

Next terms involving the input fourth cumulant

At the next connected-polynomial order, in addition to the F terms, the pieces linear in K4 and C are

$$6\sum_j c_j q_j K_4(j,j,u,u) + 6\sum_{jk} b_j b_k C_{jk} K_4(j,k,u,u) + 12\sum_{jk} b_j b_k q_j K_4(j,k,k,u).$$

For K4=Sym(G⊗ηI), put s=‖u‖², h=Gu and a=uᵀGu. The required slices are

$$K_4(j,j,u,u) = \tfrac{\eta}{6}(G_{jj}s + 4u_j h_j + a),$$

$$K_4(j,k,u,u) = \tfrac{\eta}{6}(G_{jk}s + 2h_j u_k + 2h_k u_j + \delta_{jk} a),$$

$$K_4(j,k,k,u) = \tfrac{\eta}{6}(2G_{jk}u_k + h_j + G_{kk}u_j + 2\delta_{jk}h_k).$$

These require two extra query products, C(b∘u) and (C∘G)b, plus reductions; G(b∘u) reuses the first because G=λC+diagonal. Cost: **2m/n u/layer**, plus O(n²+nm), rather than a fourth-order tensor. State: one extra n×n Hadamard matrix. This is a priced algebraic extension, not a validated accuracy improvement. K3² terms occur at the same polynomial order and are not thereby included.

Falsifier/end-to-end. Add this extension to the one-layer pair-subtracted diagnostic only: <0.7 u in principal products for 350 queries, plus slice matching. Reject it if the target residual worsens. Deploy only if recomputing all matching terms and the full end-to-end score pays for the added cost; retaining selected K4×C terms has no general error guarantee.

Appendix B / Explicit O(n²) pair matching

Here a=d₁, b=d₂, c=d₃ and s=diag(C); these b,c are unweighted local coefficients. For i≠j the Gaussian-star/path part of the quartic has the following slices:

$$H^G_{iiii} = 4c_i s_i^3 a_i^3 + 12 b_i^2 s_i^3 a_i^2,$$

$$H^G_{iiij} = 3c_i s_i^2 a_i^2 C_{ij} a_j + c_j C_{ij}^3 a_i^3 + 6 b_i^2 s_i^2 a_i C_{ij} a_j + 6 b_i b_j C_{ij}^2 s_i a_i^2,$$

$$H^G_{iijj} = 2C_{ij}^2 (c_i s_i a_i a_j^2 + c_j s_j a_j a_i^2 + b_i^2 s_i a_j^2 + b_j^2 s_j a_i^2) + 4 b_i b_j C_{ij} a_i a_j (s_i s_j + C_{ij}^2).$$

Add the harmonic-input pieces ηg_i a_i⁴ on the diagonal, (η/2)G_ij a_i³a_j for (3,1), and (η/6)(g_i+g_j)a_i²a_j² for (2,2). All operations are entrywise outer scalings and Hadamard powers; the diagonal is handled separately. These explicit expressions are the O(n²) construction used in the product ledger.

Matching the K3×C extension also stays O(n²)

Write D_ij=K3(i,i,j), d_i=K3(i,i,i). On a two-coordinate query w=xe_i+ye_j, define

$$B_i = b_i x (s_i a_i x + C_{ij} a_j y), \qquad B_j = b_j y (C_{ij} a_i x + s_j a_j y).$$

The added polynomial restricted to this pair is

$$12 B_i (d_i a_i^2 x^2 + 2 D_{ij} a_i a_j xy + D_{ji} a_j^2 y^2) + 12 B_j (D_{ij} a_i^2 x^2 + 2 D_{ji} a_i a_j xy + d_j a_j^2 y^2).$$

Take the x⁴ coefficient for its diagonal, the x³y coefficient divided by four for (3,1), and the x²y² coefficient divided by six for (2,2). Expand these scalar polynomials entrywise across pairs; do not call a dense F evaluator n² times. Hence the expensive off-slice directional query adds no further dense products for pair matching.

For the diagonal-only estimator, only H_iiii and H_iijj are required. The oracle diagnostic uses H_iiij as well to remove all known pair content. For deployment with κ31 feedback, use the full gradient construction on page 5; a current-coordinate pair slice alone is not the next-layer rotated slice.

Appendix C / Minimal NumPy diagnostics

All arrays below use ordinary cumulants, not the reference program's rescaled Wick buffers. W has columns as output queries. The first routine costs O(n²); the second costs two n×n by n×m products. Matching the known slices is specified on page 5.

```
import numpy as np

def teacher_cov_noise(Cpost, K22post, N=60_000_000):
    v = np.diag(Cpost)
    # Leading variance, not a noise-corrected residual by itself.
    return (K22post + v[:,None]*v[None,:] + Cpost*Cpost)/N

def missing_cov_terms(C, d2, d3, d4, K31):
    C = C.copy()
    np.fill_diagonal(C, 0)
    C2 = C*C
    out = (C2*C)*(d3[:,None]*d3[None,:])/6
    out += (C2*C2)*(d4[:,None]*d4[None,:])/24
    B = d4[:,None]*K31*d2[None,:]
    out += C*(B+B.T)/6
    return out

def project_variance(E, W):
    return np.sum(W*(E@W), axis=0)

def quartic_kernel(C, W, d1, d2, d3, g, lam, metric):
    u = d1[:,None]*W
    b = d2[:,None]*W
    c = d3[:,None]*W
    q = C@u
    t = b*q
    r = C@t
    h = lam*q + (g-lam*np.diag(C))[:,None]*u
    f = metric*np.sum(u*u,axis=0)*np.sum(u*h,axis=0)
    f += 4*np.sum(c*q*q*q,axis=0)
    f += 12*np.sum(t*r,axis=0)
    return f
```

Use the actual next W and the chain-chosen sensitivity mask. Compare projected errors with and without the correction, weighted by the squared local mean derivative. The leading derivatives are $\partial m/\partial v=\varphi(\alpha)/(2\sigma)$, $\partial m/\partial \kappa_3=-\alpha\varphi(\alpha)/(6\sigma^2)$, and $\partial m/\partial \kappa_4=(\alpha^2-1)\varphi(\alpha)/(24\sigma^3)$. Prefer the exact derivatives of your implemented mean program when ranking neurons.

Checks actually executed for this report

Exact Gauss-Hermite integration of small polynomial examples verified the star, path and quadratic-ring coefficients to $<3\times10^{-16}$. Finite differences verified the quartic gradient to 1.8×10^{-11} relative error. Explicit small tensors verified the pair-slice formulas and factored directional K3 query to $<6\times10^{-15}$. These are algebra checks, not tests on your network or Monte Carlo teacher.

Sources and reproducibility

The cited sources support the source-code audit and the general derivative/connected-cumulant tools. The network-specific contractions, error budgets, cost ledgers and falsifiers are derived here. No paper is cited as a validated solution of this benchmark.

[1] 504aldo, public V29 implementation: TERM_SPECS[(1,1)], wk431 additions and covariance symmetrization. Local source inspection; audit your active branch before applying the patch.

GitHub / estimator_v29.py / pinned source 18c17e2d

[2] Felix Voigtlaender, A general version of Price's theorem. Supports Gaussian expectation derivatives for nonsmooth functions through distributional derivatives; the particular correction formulas here are derived in this report.

https://arxiv.org/abs/1710.03576

[3] Jani Lukkarinen and Matteo Marcozzi, Wick polynomials and time-evolution of cumulants. Primary reference for connected cumulant expansions and removal of internal contractions; no claim that it solves this network benchmark.

https://arxiv.org/abs/1503.05851

Scope of verification

The Gaussian covariance terms and wk431 extension were checked in the available public V29 source. A small independent NumPy program checked the quartic polynomial, its gradient, local pair slices and the factored K3 contraction. Its largest reported pair-matching discrepancy was 1.1×10^{-14}. No teacher arrays, teacher sample blocks or raw input/output sample pairs were supplied for Round 7. Therefore every proposed network-accuracy improvement remains a falsifiable hypothesis.

Practical implementation boundaries

All persistent state proposed here is $O(n^2)$ in addition to the existing K3 source representation. $O(nm)$ query work can be blocked to stay well below the memory cap. Dense-product estimates include the stated Strassen cost only for square products; shared source-basis and rectangular query products use explicit classical counts. Gaussian CDF/PDF and Wick coefficients are reused from the existing closure. Additional source thin legs, conversion conventions and any fallback factorizations must be counted by the shipping meter.

A first falsifier is a local negative test, not a proof of final accuracy. A local pass must be followed by the prescribed end-to-end comparison; a failure closes that approximation without reviving previously failed rank, age-window or sampling proposals.